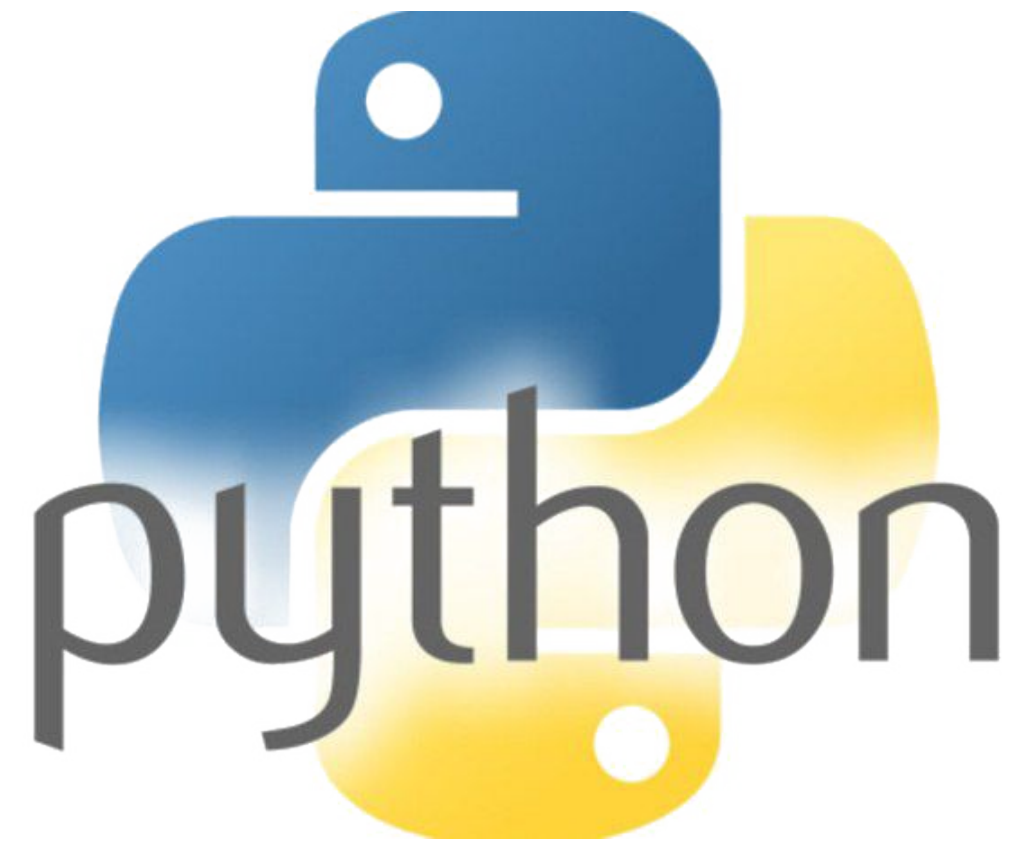




Index

**Applica
-tions**

What is
Python..?

Basic
History

Key-word
s

Features
Of
Pyhton



What is Python ..?



: - Python is a general purpose, high level, and interpreted programming language. It supports Object Oriented programming approach to develop applications. It is simple and easy to learn and provides lots of high-level data structures.

: - Python is *easy to learn* yet powerful and versatile scripting language, which makes it attractive for Application Development..

Example :

Python program to add two numbers :

Input- num1 = 5

num2 = 3

sum = num1 + num2

Print("sum of {0} and {1} is {2}" .format(num1, num2, sum))

Output- sum of 5 and 3 is 8

History of Python

“Python laid its foundation in the late **1980s.**”

“The implementation of Python was started in the **December 1989** by **Guido Van Rossum** at CWI in Netherland”

“In **February 1991**, van Rossum published the code (**labeled version 0.9.0**) to alt.sources.”

In **1994**, Python 1.0 was released with new features like:
lambda, map, filter, and reduce.

Python 2.0 added new features like: **list comprehensions, garbage collection system.**



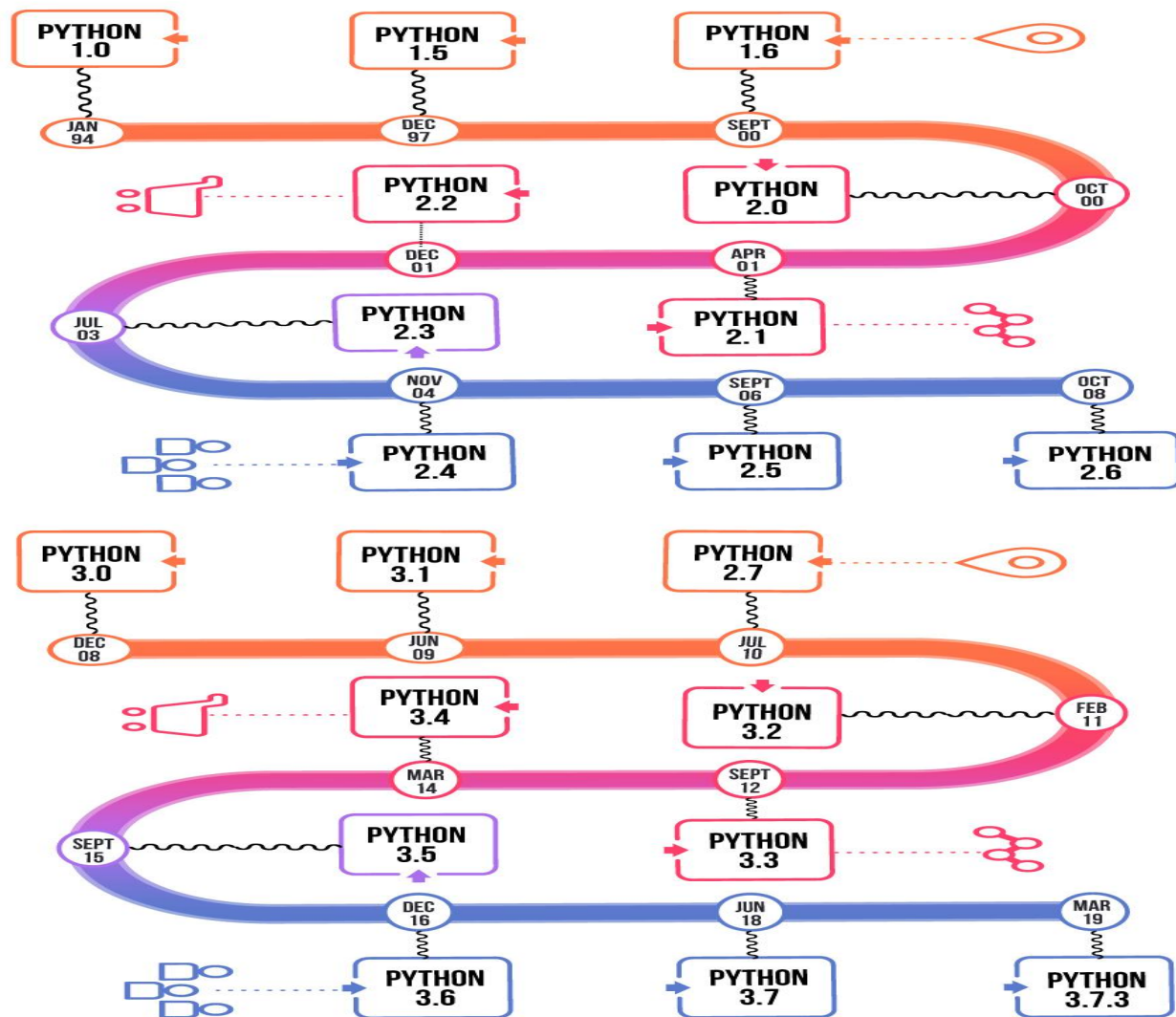
History of Python

On December 3, 2008, Python 3.0 (also called "Py3K") was released. It was designed to rectify fundamental flaw of the language.

ABC programming language is said to be the predecessor of Python language which **was capable of Exception Handling and interfacing with Amoeba Operating System.**

Python is influenced by following programming Languages :
ABC language
Modula-3





Features of Python

Easy

“Python is easy to learn and use. It is developer-friendly and high level programming language.”

Expressive Language

“Python language is more expressive means that it is more understandable and readable.”

Interpreted Language

“Python is an interpreted language i.e. interpreter executes the code line by line at a time. This makes debugging easy and thus suitable for beginners.”



Features of Python

Extensible

“It implies that other languages such as C/C++ can be used to compile the code and thus it can be used further in our python code.”

Large Standard Library

“Python has a large and broad library and provides rich set of module and functions for rapid application development..”

GUI Programming Support

“Graphical user interfaces can be developed using Python..”



Features of Python

Cross-platform Language

“Python can run equally on different platforms such as Windows, Linux, Unix and Macintosh etc. So, we can say that Python is a portable language.”

Free and Open Source

“Python language is freely available at official web address.The source-code is also available. Therefore it is open source.”

Object-Oriented Language

“Python supports object oriented language and concepts of classes and objects come into existence..”



Features of Python

Integrated

“It can be easily integrated with languages like C, C++, JAVA etc.”

High-Level Language

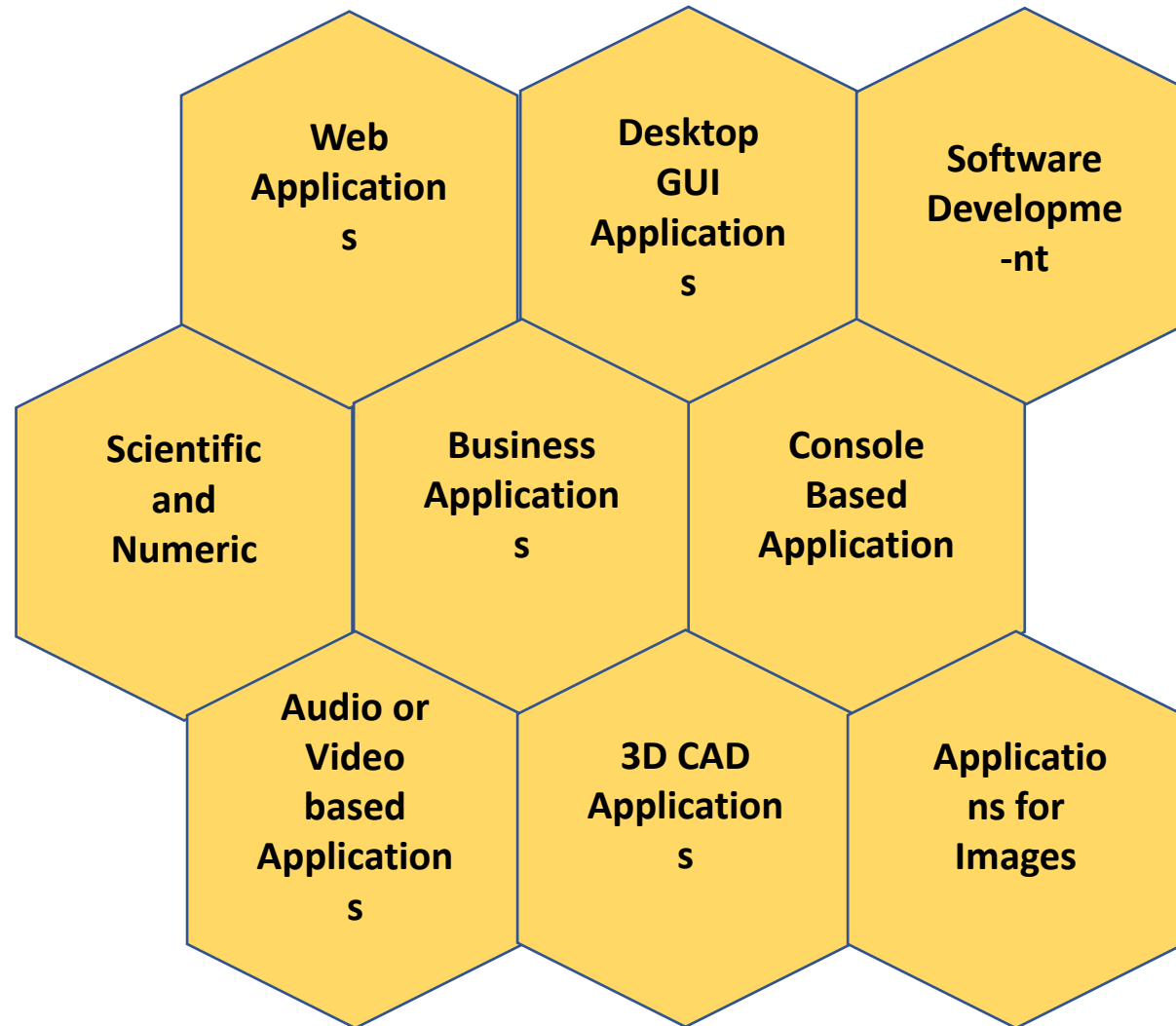
“When we write programs in python, we do not need to remember the system architecture, nor do we need to manage the memory..”

Dynamically Typed Language

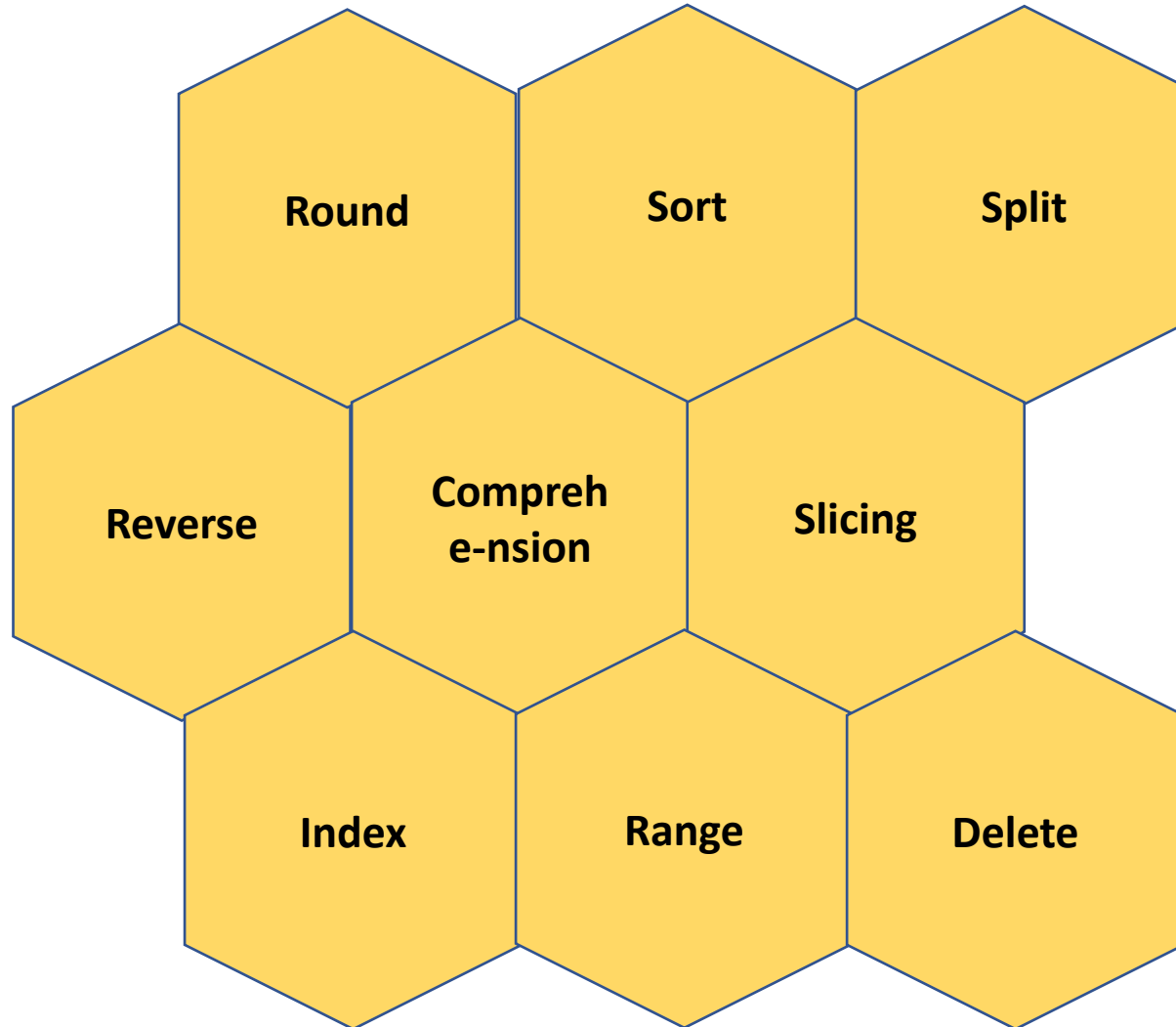
“That means the type (for example- int, double, long etc) for a variable is decided at run time not in advance. because of this feature we don't need to specify the type of variable..”



Applications



Keywords



Round

Example :

Input –
`x = round(5.76543, 2)`
`print(x)`

Output –
`5.77`

Sort

Example :

Input –

#List of integers
`Numbers = [1,3,4,2]`

#Sorting list of integers
`Numbers .Sort()`

`print(Numbers)`

Output –

`[1,2,3,4]`

Split

Example :

Input –

`text = 'geeks for
geeks'`

Splits at space

`print(text.split())`

Output –

`['geeks', 'for', 'geeks']`

Reverse

Example :

Input –

#a list of numbers
`list1 = [1, 2, 3, 4, 1, 2, 6]`
`list1.reverse()`
`print(list1)`

Output –

`[6, 2, 1, 4, 3, 2, 1]`

Slicing

Example :

Input –

```
#list[start : end : step]
list1 =
[1,2,3,4,5,6,7,8]
print("list: "list1)
print("After Slicing")
print(list1[1:4:1])
```

Output –

```
List : [1,2,3,4,5,6,7,8]
After Slicing
[2,3,4]
```

Index

Example :

Input –

```
#list.index() method
list1 = [1, 2, 3, 4, 1, 1, 1, 4,
5]
# Will print the index of '4' in
list1
```

```
print(list1.index(4))
```

Output –

3

Range

Example :

Input –

```
#printing a number
for i in range(10):
print(i, end = " ")
print()
```

Output –

0 1 2 3 4 5 6 7 8 9

Delete

Example :

Input –

```
list = [2, 1, 3, 5, 4, 3, 8]
# deletes 3,5,4
del lis[2 : 5]
```

Output –

List elements after
deleting are : 2 1 3 8

